

Now, define the current through the circuit with respect to the output signal using Ohm's Law

$$i_a(s) = \frac{V_o(s)}{\frac{1}{Cs}} = CsV_o(s)$$

the KVL equation simplifies to

$$-V_a(s) + RCsV_o(s) + LCs^2V_o(s) + V_o(s) = 0$$

Since a transfer function is a ratio of output to input, we need to isolate the I/O signals

$$V_a(s) = RCsV_o(s) + LCs^2V_o(s) + V_o(s)$$

$$V_a(s) = (RCs + LCs^2 + 1)V_o(s)$$

$$\frac{V_o(s)}{V_a(s)} = \frac{1}{RCs + LCs^2 + 1}$$

Finally, in proper form

$$\frac{V_o(s)}{V_a(s)} = \frac{1}{LCs^2 + RCs + 1}$$

Note: "Proper form" implies the numerator and denominator are written in descending order polynomial form.

5.2 Mechanical Diagram

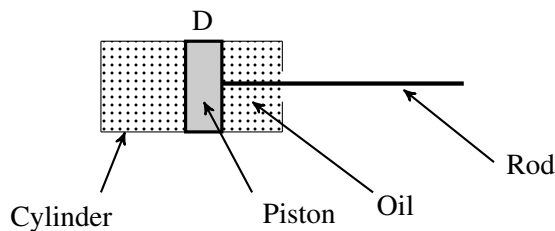
In electrical engineering we use a circuit diagram to find the Laplace domain equation, then the transfer function, block diagram, etc. In mechanical engineering we use a free body diagram (also called a force diagram) to find the Laplace domain equation, transfer function, etc. For dynamic mechanical systems, there are several common devices to consider.

Damper

The damper absorbs energy and dissipates it as heat, it does not store kinetic or potential energy.

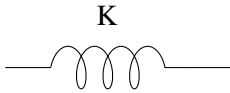
As constructed, it's a cylinder, piston, rod, and oil. The cylinder is filled with oil and the leaky piston – which is connected to the rod.

Relative motion between the rod and the cylinder forces oil to move from one side of the piston to the other. The amount of energy that is dissipated is a function of the physical characteristics, such as, the oil viscosity and the leakiness of the piston. The mathematical model uses a damper coefficient D to represent the characteristics of the device.



Spring

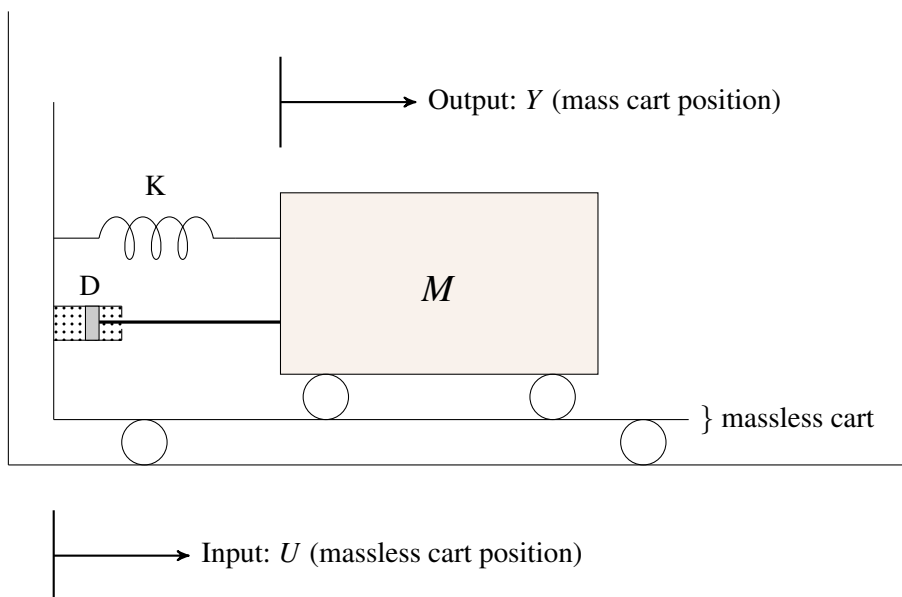
Every spring has a natural relaxed position called the free length. Compress a spring and it opposes the compression. Stretch a spring it opposes the stretching. The mathematical models uses a spring constant K .



Mechanical systems utilize a free body diagram much like electrical systems use electrical circuits.

Free Body Diagram

Given a free body diagram composed of a cart with mass M sitting on a massless cart. The two carts are connected to each other with a spring and a damper. The input to the system is the position of the massless cart and the output is the position of the mass cart.



To find the open-loop transfer function from the free body diagram, we start with Newton's second law

$$\sum F = ma \quad (5.1)$$

where $\sum F$ is the summation of all of the forces acting on the system, m is the mass of the mass cart and a is the acceleration of the mass cart. Since $y(t)$ is defined as the position of the mass cart, the acceleration of the mass cart is the second derivative of the cart's position

$$a = \ddot{y}(t)$$

and

$$\sum F = F_D + F_S$$

where F_D is the damper force

$$F_D = -D(\dot{y}(t) - \dot{u}(t))$$

and F_S is the spring force

$$F_S = -K(y(t) - u(t))$$

Newton's second law for the free body diagram is

$$m\ddot{y}(t) = -D(\dot{y}(t) - \dot{u}(t)) - K(y(t) - u(t))$$

$$m\ddot{y}(t) = -D\dot{y}(t) + D\dot{u}(t) - Ky(t) + Ku(t)$$

transform into the Laplace domain

$$ms^2Y(s) = -DsY(s) + DsU(s) - KY(s) + KU(s)$$

isolate the I/O signals

$$ms^2Y(s) + DsY(s) + KY(s) = DsU(s) + KU(s)$$

and factor them

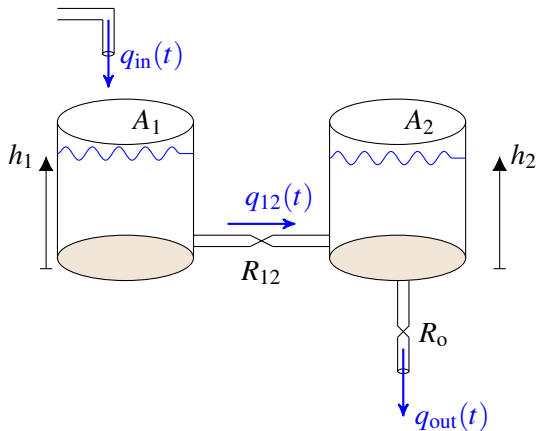
$$(ms^2 + Ds + K)Y(s) = (Ds + K)U(s)$$

Finally, write the open-loop transfer function

$$\frac{Y(s)}{U(s)} = \frac{Ds + K}{ms^2 + Ds + K}$$

5.3 Two Tank Fluid Flow Describing Diagram

Given the describing diagram for a two tank fluid flow problem



From the diagram, determine the fluid flow from input to output and construct an open-loop transfer function. First, define the differential equations. The volume of fluid in the first tank is

$$V_1(t) = A_1 \cdot h_1(t)$$

however, there is fluid flowing into the tank, $q_{in}(t)$ and fluid flowing out of the tank, $q_{12}(t)$; so we consider the rate of change of the fluid in the first tank.

$$q_{in}(t) - q_{12}(t) = \frac{dV_1(t)}{dt}$$

or written with h_1 as the dynamic variable

$$q_{in}(t) - q_{12}(t) = A_1 \cdot \frac{dh_1(t)}{dt} \quad (5.2)$$

The flow from tank 1 to tank 2 is a function of the heights of the tanks and the restriction of the flow tube.

$$q_{12}(t) = \frac{h_1(t) - h_2(t)}{R_{12}} \quad (5.3)$$

where $1 \leq R_{12} < \infty$, the larger R_{12} the greater the restriction on the flow rate.

Similar to the fluid flow in tank 1, the fluid flow and change of volume of tank 2 is

$$q_{12}(t) - q_{out}(t) = A_2 \cdot \frac{dh_2(t)}{dt} \quad (5.4)$$

Finally, the output equation is simply

$$q_{out}(t) = \frac{h_2(t)}{R_o} \quad (5.5)$$

where $1 \leq R_o < \infty$, the larger R_o the greater the restriction on the flow rate.

Now that we have defined the system from input to output we need to convert (5.2) to (5.5) into the Laplace domain equivalent equations

Starting with (5.2)

$$Q_{in}(s) - Q_{12}(s) = A_1 s H_1(s)$$

now, we rewrite it with the input $Q_{in}(s)$ and the output $H_1(s)$

$$H_1(s) = \frac{Q_{in}(s) - Q_{12}(s)}{A_1 s} \quad (5.6)$$

Moving to (5.3)

$$Q_{12}(s) = \frac{H_1(s) - H_2(s)}{R_{12}} \quad (5.7)$$

Next, (5.4)

$$Q_{in}(s) - Q_{12}(s) = A_2 s H_2(s)$$

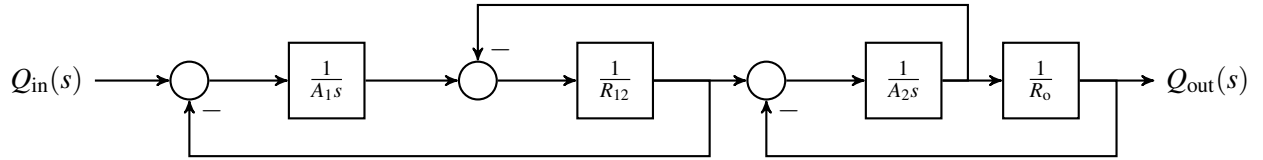
Now, we rewrite the equation with the input $Q_{in}(s)$ and the output $H_1(s)$

$$H_2(s) = \frac{Q_{12}(s) - Q_{out}(s)}{A_2 s} \quad (5.8)$$

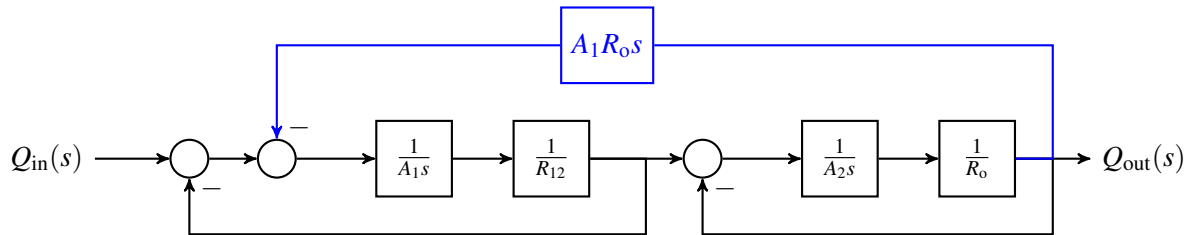
Finally (5.5)

$$Q_{out}(s) = \frac{H_2(s)}{R_o} \quad (5.9)$$

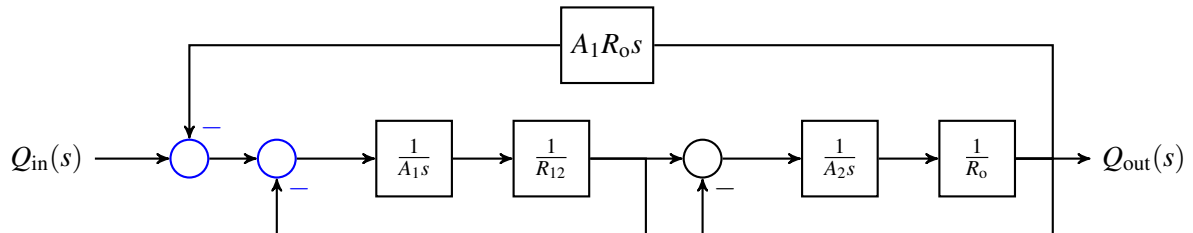
Now, we use the Laplace domain equations to construct the block diagram of the system



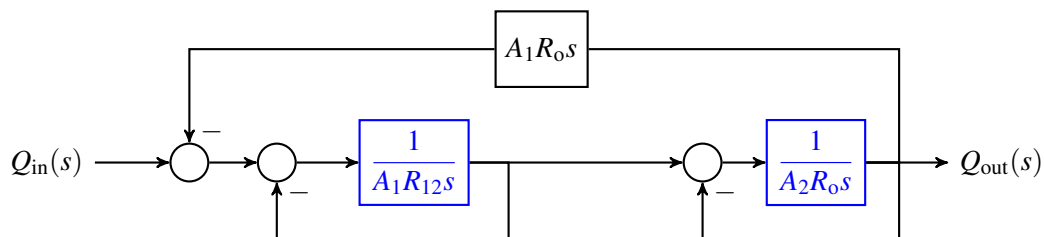
Next, we need to simplify the block diagram down to a transfer function. First move the branch from in front of the $\frac{1}{R_o}$ to after it and move the summation junction after $\frac{1}{A_1 s}$ to before it



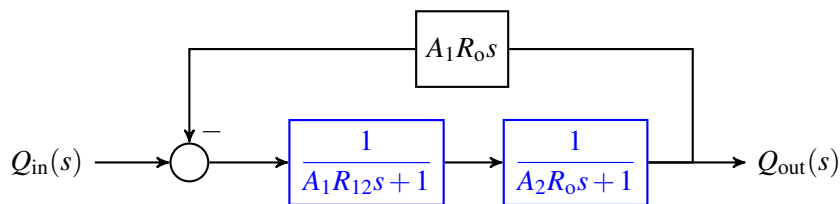
Change the order of the first two summation junctions



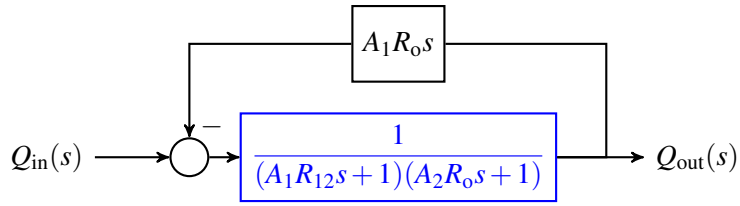
Combine cascaded blocks in the forward path



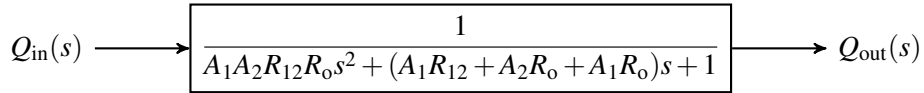
Next, eliminate the unity feedback loops



Combine the two remaining cascaded blocks in the forward path



Finally, eliminate the remaining feedback loop and reduce the diagram to produce a single open-loop transfer function block



Of course, the single open-loop block diagram reveals the input/output transfer function

$$\frac{Q_{\text{out}}(s)}{Q_{\text{in}}(s)} = \frac{1}{A_1 A_2 R_{12} R_o s^2 + (A_1 R_{12} + A_2 R_o + A_1 R_o) s + 1}$$